

# SOLUTIONS GUIDE

Total: 62 points

Time: 60 minutes

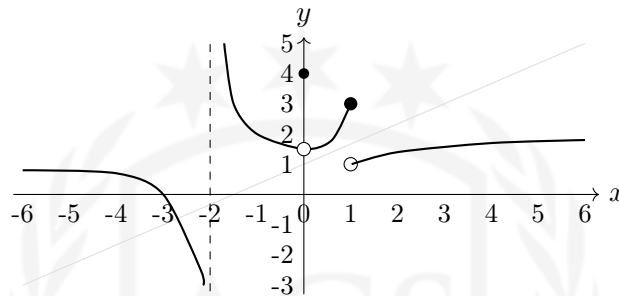
## Part I: No Calculator

1. (3 points) Use the table to evaluate the limit. Explain briefly how the table supports your answer.

|        |      |       |           |       |      |
|--------|------|-------|-----------|-------|------|
| $x$    | 1.9  | 1.99  | 2         | 2.01  | 2.1  |
| $f(x)$ | 3.91 | 3.991 | undefined | 4.009 | 4.09 |

 $\lim_{x \rightarrow 2} f(x) = \underline{\hspace{2cm}}$ 

2. Use the graph of  $f(x)$  below to answer the following questions.



- (a) (1 point)  $\lim_{x \rightarrow \infty} f(x) =$   
(b) (1 point)  $\lim_{x \rightarrow -2^+} f(x) =$   
(c) (1 point)  $\lim_{x \rightarrow -2^-} f(x) =$   
(d) (1 point)  $\lim_{x \rightarrow -2} f(x) =$   
(e) (1 point)  $\lim_{x \rightarrow 0} f(x) =$   
(f) (1 point)  $\lim_{x \rightarrow 1^-} f(x) =$   
(g) (2 points) Is  $f(x)$  continuous at  $x = 0$ ? Use the definition of continuity to explain why or why not.

**Solution:** False, we would have to jump from the hole to point  $(0, 4)$  which makes it not continuous.

3. Let

$$\begin{cases} f(x) = x^2 - 4, & x \leq 2, \\ 3x - 6, & x > 2. \end{cases}$$

- (a) (3 points) Find  $\lim_{x \rightarrow 2^-} f(x)$  and  $\lim_{x \rightarrow 2^+} f(x)$ .

**Solution:**  $\lim_{x \rightarrow 2^-} f(x) = 0$  and  $\lim_{x \rightarrow 2^+} f(x) = 0$

- (b) (2 points) Is  $f$  continuous at  $x = 2$ ? Use the definition of continuity to justify your answer.

**Solution:** Yes, because  $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = f(2) = 0$

4. Suppose  $\lim_{x \rightarrow c} f(x) = 5$  and  $\lim_{x \rightarrow c} g(x) = -2$ . Find each limit.

- (a) (1 point)  $\lim_{x \rightarrow c} [2f(x) - 3g(x)]$

**Solution:**  $\lim_{x \rightarrow c} [2f(x) - 3g(x)] = 2(5) - 3(-2) = 10 + 6 = 16$

- (b) (1 point)  $\lim_{x \rightarrow c} \frac{g(x)}{f(x)}$

**Solution:**  $\lim_{x \rightarrow c} \frac{g(x)}{f(x)} = \frac{-2}{5} = -\frac{2}{5}$

- (c) (1 point)  $\lim_{x \rightarrow c} [f(x)g(x)]$

**Solution:**  $\lim_{x \rightarrow c} [f(x)g(x)] = (5)(-2) = -10$

5. Find each limit algebraically, if it exists.

- (a) (2 points)  $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$

**Solution:**  $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4} = \lim_{x \rightarrow 4} \frac{(x - 4)(x + 4)}{x - 4} = \lim_{x \rightarrow 4} (x + 4) = 8$

- (b) (2 points)  $\lim_{x \rightarrow \infty} \frac{6x^2 - x + 1}{2x^2 + 5}$

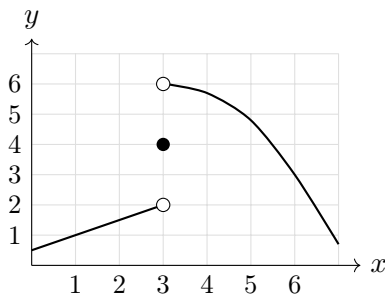
**Solution:**  $\lim_{x \rightarrow \infty} \frac{6x^2 - x + 1}{2x^2 + 5} = \lim_{x \rightarrow \infty} \frac{6 - \frac{1}{x} + \frac{1}{x^2}}{2 + \frac{5}{x^2}} = \frac{6}{2} = 3$

6. (3 points) Find the  $x$ -values, if any, at which  $f(x)$  is not continuous.

$$f(x) = \begin{cases} x^2 - 2, & x \leq 1, \\ \frac{x^2 - 4}{x - 2}, & x > 1. \end{cases}$$

**Solution:**  $f(x)$  is not continuous at  $x = 2$  because the limit from the left does not equal the limit from the right.

7. (2 points) Which of the following expressions equals 2 for the graph shown below?



- (A)  $f(3)$  (C)  $\lim_{x \rightarrow 3^+} f(x)$   
 (B)  $\lim_{x \rightarrow 3^-} f(x)$  (D)  $\lim_{x \rightarrow 3} f(x)$

**Solution:** The correct answer is (B)  $\lim_{x \rightarrow 3^-} f(x)$  because the limit from the left approaches 2.

8. (2 points) For  $x < 0$ , the line  $y = 3$  is an asymptote of the graph of a function  $f$ . Which statement must be true?

- (A)  $f(x) \neq 3$  for  $x < 0$  (C)  $\lim_{x \rightarrow \infty} f(x) = 3$   
 (B)  $\lim_{x \rightarrow -\infty} f(x) = 3$  (D)  $\lim_{x \rightarrow 3} f(x) = \infty$

**Solution:** The correct answer is (B)  $\lim_{x \rightarrow -\infty} f(x) = 3$  because an asymptote indicates that the function approaches that value as  $x$  goes to negative infinity.

9. (2 points) If  $f$  is continuous on the closed interval  $[2, 5]$ ,  $f(2) = -3$ , and  $f(5) = 4$ , which statement must be true?

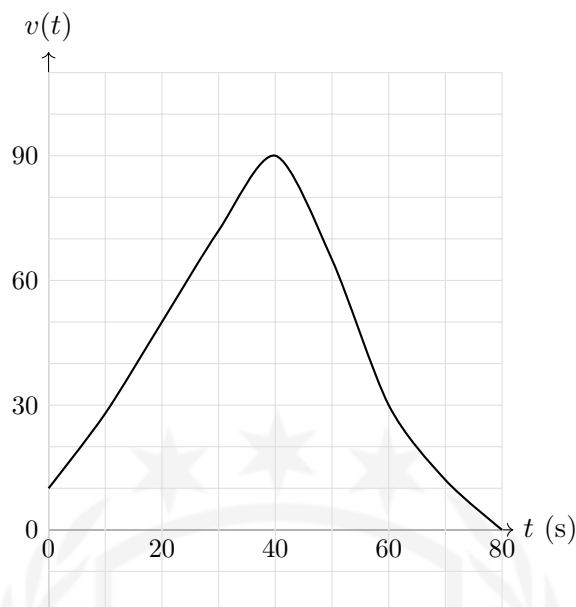
- (A)  $f(c) = 2$  for some  $c \in (-3, 4)$  (C)  $f(c) = 0$  for some  $c \in (-3, 4)$   
 (B)  $f(c) \neq 1$  for every  $c \in (2, 5)$  (D)  $f(c) = 2$  for some  $c \in (2, 5)$

**Solution:** The correct answer is (D)  $f(c) = 2$  for some  $c \in (2, 5)$  because by the Intermediate Value Theorem, since  $f(2) = -3$  and  $f(5) = 4$ , there must be some value  $c$  in the interval  $(2, 5)$  where  $f(c) = 2$ .



## Part II: Calculator Allowed

10. The velocity  $v(t)$ , in feet per second, of a car is represented by the graph below.



- (a) (2 points) Estimate the instantaneous rate of change of  $v$  at  $t = 60$  by drawing a tangent line and using a difference quotient.

**Solution:** The instantaneous rate of change at  $t = 60$  can be estimated by drawing a tangent line at that point and calculating the slope. For example, if the tangent line passes through points  $(60, 30)$  and  $(70, 12)$ , the slope would be  $\frac{12-30}{70-60} = \frac{-18}{10} = -1.8$  ft/s<sup>2</sup>.

- (b) (1 point) What does the rate of change in part (a) represent in this context?

**Solution:** The rate of change represents the acceleration of the car at  $t = 60$  seconds, indicating how quickly the velocity is changing at that moment.

- (c) (2 points) Estimate the distance traveled from  $t = 20$  to  $t = 60$  by counting geometric areas.

**Solution:** The distance traveled can be estimated by calculating the area under the velocity curve from  $t = 20$  to  $t = 60$ . This can be done by approximating the area using geometric shapes (triangles and rectangles) or using numerical integration methods.

- (d) (3 points) Estimate the total distance traveled from  $t = 0$  to  $t = 80$  by using the Trapezoid Rule with four subintervals.

**Solution:** To estimate the total distance traveled from  $t = 0$  to  $t = 80$  using the Trapezoid Rule with four subintervals, divide the interval into four equal parts:  $[0,20]$ ,  $[20,40]$ ,  $[40,60]$ , and  $[60,80]$ . Calculate the area of each trapezoid formed by the velocity values at the endpoints of each subinterval and sum them up to get the total distance.

11. (5 points) For  $s(t) = -t^2 + 12t + 5$ , find the average velocity on the interval  $[2, 2.1]$ . Then state the instantaneous velocity at  $t = 2$  by taking the appropriate limit.

**Solution:** The average velocity on the interval  $[2, 2.1]$  is calculated as:

$$\text{Average velocity} = \frac{s(2.1) - s(2)}{2.1 - 2} = \frac{(-2.1^2 + 12(2.1) + 5) - (-2^2 + 12(2) + 5)}{0.1}$$

Calculate  $s(2)$  and  $s(2.1)$ :

$$s(2) = -4 + 24 + 5 = 25, \quad s(2.1) = -4.41 + 25.2 + 5 = 25.79$$

Thus,

$$\text{Average velocity} = \frac{25.79 - 25}{0.1} = \frac{0.79}{0.1} = 7.9 \text{ m/s}$$

The instantaneous velocity at  $t = 2$  is found by taking the derivative of  $s(t)$ :

$$v(t) = s'(t) = -2t + 12$$

Evaluating at  $t = 2$ :

$$v(2) = -4 + 12 = 8 \text{ m/s}$$

12. A particle moves along a line with position

$$s(t) = 0.4t^3 - 3t^2 + 8t + 2$$

for  $0 \leq t \leq 6$ , where  $s$  is measured in meters and  $t$  in seconds.

- (a) (2 points) Find the particle's average velocity on  $2 \leq t \leq 5$ .

**Solution:** The average velocity is given by:

$$\text{Average velocity} = \frac{s(5) - s(2)}{5 - 2}$$

Calculate  $s(2)$  and  $s(5)$ :

$$s(2) = 0.4(8) - 3(4) + 8(2) + 2 = 3.2 - 12 + 16 + 2 = 9.2$$

$$s(5) = 0.4(125) - 3(25) + 8(5) + 2 = 50 - 75 + 40 + 2 = 17$$

Thus,

$$\text{Average velocity} = \frac{17 - 9.2}{3} = \frac{7.8}{3} = 2.6 \text{ m/s}$$

- (b) (2 points) Use the calculator to find the instantaneous velocity at  $t = 3$ , and interpret its sign.

**Solution:** The instantaneous velocity is given by the derivative of  $s(t)$ :

$$v(t) = s'(t) = 1.2t^2 - 6t + 8$$

Evaluating at  $t = 3$ :

$$v(3) = 1.2(9) - 6(3) + 8 = 10.8 - 18 + 8 = 0.8 \text{ m/s}$$

The positive sign indicates that the particle is moving in the positive direction along the line at  $t = 3$ .

13. The graph of  $f$  is modeled by

$$f(x) = \frac{x^2 - 5x + 6}{x - 2}.$$

- (a) (2 points) Use algebra to identify any discontinuity(s).

**Solution:** The function  $f(x)$  can be factored as:

$$f(x) = \frac{(x - 2)(x - 3)}{x - 2}$$

For  $x \neq 2$ , this simplifies to  $f(x) = x - 3$ . However, at  $x = 2$ , the function is undefined, indicating a discontinuity at  $x = 2$ .

- (b) (2 points) Use a graphing calculator to display the graph near the discontinuity and state the value of  $\lim_{x \rightarrow 2} f(x)$ .

**Solution:** Using a graphing calculator, we can see that as  $x$  approaches 2 from either side, the function values approach 1. Therefore,

$$\lim_{x \rightarrow 2} f(x) = 1.$$

- (c) (1 point) Find the horizontal asymptote, if one exists.

**Solution:** As  $x$  approaches infinity, the function behaves like  $f(x) = x - 3$ , which does not approach a finite value. Therefore, there is no horizontal asymptote.

14. (4 points) Find  $a$  and  $b$  so that the function is continuous for all real  $x$ :

$$f(x) = \begin{cases} 2x + a, & x < -1, \\ ax + b, & -1 \leq x \leq 3, \\ 4x - b, & x > 3. \end{cases}$$

Show the two equations you use and solve the resulting system.

**Solution:** To ensure continuity at  $x = -1$ :

$$\lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^+} f(x)$$

$$2(-1) + a = a(-1) + b \implies -2 + a = -a + b \implies 2a - b = 2 \quad (1)$$

To ensure continuity at  $x = 3$ :

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x)$$

$$a(3) + b = 4(3) - b \implies 3a + b = 12 - b \implies 3a + 2b = 12 \quad (2)$$

Solving the system of equations (1) and (2):

$$\begin{cases} 2a - b = 2 \\ 3a + 2b = 12 \end{cases}$$

From equation (1):  $b = 2a - 2$  Substituting into equation (2):

$$3a + 2(2a - 2) = 12 \implies 3a + 4a - 4 = 12 \implies 7a = 16 \implies a = \frac{16}{7}$$

Then,  $b = 2(\frac{16}{7}) - 2 = \frac{32}{7} - 2 = \frac{32-14}{7} = \frac{18}{7}$  Therefore,  $a = \frac{16}{7}$  and  $b = \frac{18}{7}$ .

15. (4 points) Verify that the Intermediate Value Theorem applies to  $p(x) = x^3 - 4x + 1$  on  $[0, 2]$ . Use a calculator to locate one value of  $c$  in  $(0, 2)$  for which  $p(c) = 0$ , rounded to three decimal places.